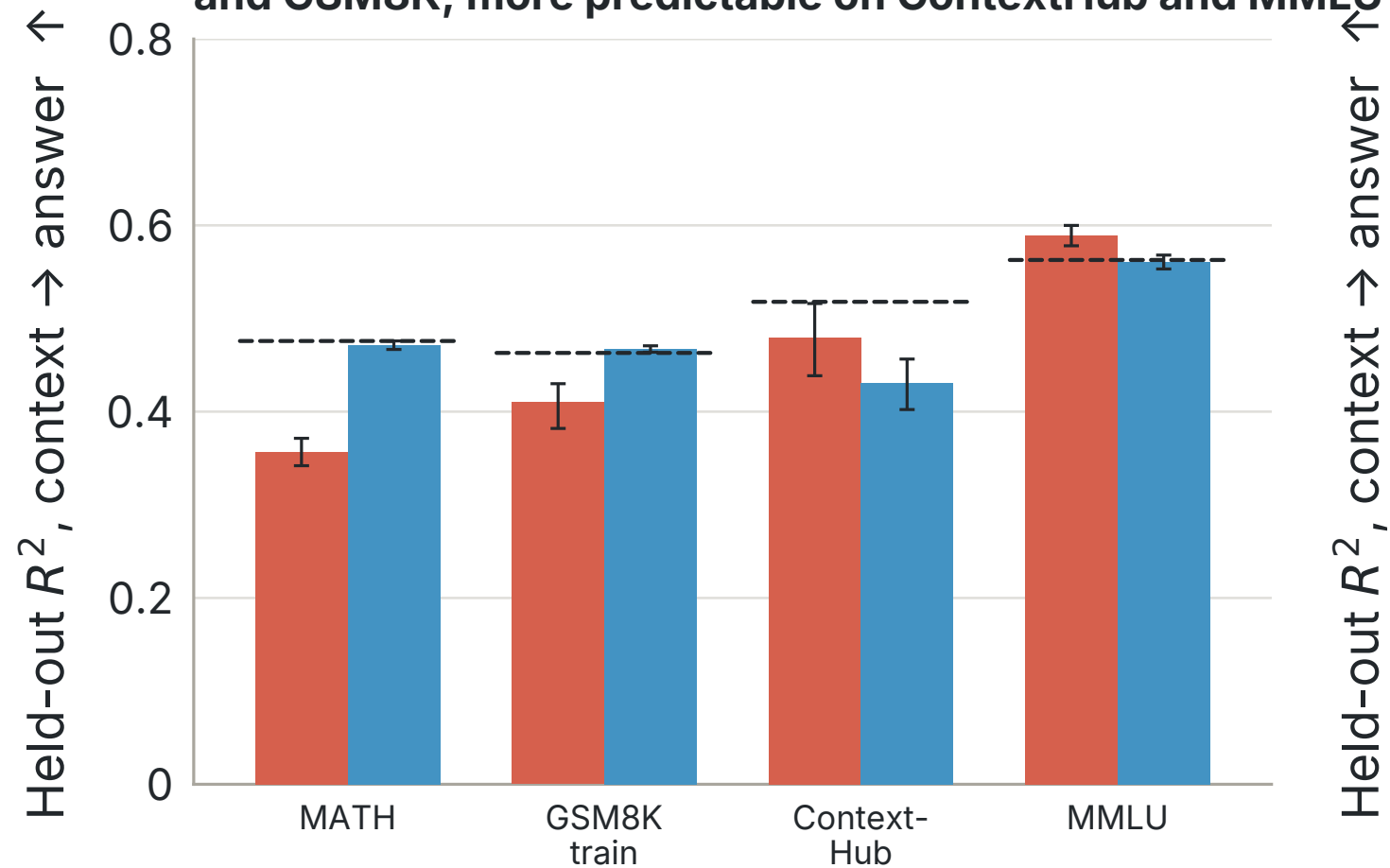


QUESTIONS, LABELED BY WHETHER REASONING WAS NEEDED

Reasoning necessary (right with reasoning, wrong without) Reasoning not needed (right both ways) ---- Whole corpus

A · OPENTHINKER3-7B, VS ITS NON-REASONING PARENT, LAYER 19

Necessary questions are less predictable on MATH and GSM8K, more predictable on ContextHub and MMLU



B · QWEN3-8B, THINKING ON VS OFF, SAME WEIGHTS, LAYER 24

Same on the thinking toggle, except that MMLU is slightly lower

